

SINOPSIS KURSUS / COURSE SYNOPSIS

1. Jabatan Biokimia / Department of Biochemistry

BCH3004	Prinsip Biokimia / Principles of Biochemistry	4(3+1)
Prasyarat: Tiada		
Kursus ini merangkumi prinsip biokimia dan penglibatan biomolekul dalam laluan utama metabolisme sistem biologi. Pelbagai struktur, ciri dan peranan biomolekul dibincangkan. Integrasi dan pengawalan metabolisme dalam sistem hidupan juga ditekankan. Penganalisan tindak balas metabolisme biomolekul juga dijalankan.		
<i>This course encompasses principles of biochemistry and involvement of biomolecules in major metabolism pathways of biological systems. Various structures, characteristics and role of biomolecules are discussed. Integration and regulation of metabolisms in living systems are also emphasized. Analyses of biomolecular metabolic reactions are also conducted.</i>		
BCH3005	Kimia Biologi / Biological Chemistry	4(3+1)
Prasyarat: Tiada		
Kursus ini merangkumi prinsip dan teknik asas kimia biologi dalam menganalisis struktur biomolekul. Konsep kimia organik dalam mengenal pasti kumpulan berfungsi struktur biomolekul ditekankan. Eksperimen menggunakan peralatan asas makmal dalam kimia biologi turut dijalankan.		
<i>This course encompasses principles and basic techniques of biological chemistry in analyzing biomolecules structures. Concept of organic chemistry in identifying functional groups of biomolecules structures are emphasized. Experimentation using basic laboratory instruments in biological chemistry are also conducted.</i>		
BCH3106	Teknik dan Pengiraan Asas dalam Biokimia / Basic Techniques and Calculations in Biochemistry	2(1+1)
Prasyarat: Tiada		
Kursus ini merangkumi teknik dan pengiraan asas dalam biokimia. Eksperimen menggunakan peralatan asas dalam biokimia dan penganalisan data turut ditekankan.		
<i>This course encompasses techniques and basic calculations in biochemistry. Experimentation using basic biochemistry instruments and data analysis are also emphasized</i>		
BCH3107	Biomolekul / Biomolecules	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi pelbagai struktur dan ciri biomolekul dalam sistem biologi. Peranan dan fungsi biomolekul seperti karbohidrat, asid amino, protein, lipid dan asid nukleik dalam sistem biologi dibincangkan.		
<i>This course encompasses various structures and characteristics of biomolecules in the biological system. Role and function of biomolecules such as carbohydrates, amino acids, proteins, lipids and nucleic acids in biological system are discussed.</i>		
BCH3108	Enzimologi / Enzymology	4(3+1)
Prasyarat: BCH3004 atau BCH3107		
Kursus ini merangkumi ciri umum enzim, faktor yang mempengaruhi keaktifan enzim dan aplikasi enzim dalam diagnostik. Faktor keaktifan dan pengawalaturan aktiviti enzim dibincangkan. Penulenan enzim daripada sistem kehidupan menggunakan pelbagai teknik penulenan turut dijalankan.		

This course encompasses general characteristics and factors affecting enzyme activity. Enzyme activation factors and regulations are discussed. Enzyme purification process from the living system using various purification techniques is also conducted.

BCH3109	Metabolisme Karbohidrat / Carbohydrate Metabolism	3(2+1)
Prasyarat: BCH3204 atau BCH3108		
Kursus ini merangkumi metabolisme karbohidrat di dalam sel. Integrasi dan mekanisme pengawalan metabolisme karbohidrat dibincangkan. Ujikaji dan penganalisaan tindak balas metabolisme karbohidrat turut dijalankan.		
<i>This course encompasses carbohydrate metabolism in cell. Integration and regulation mechanisms of carbohydrate metabolism are discussed. Experimentation and analysis of carbohydrates metabolic processes are conducted.</i>		

BCH3110	Metabolisme Protein dan Asid Nukleik / Protein and Nucleic Acid Metabolism	4(3+1)
Prasyarat: BCH3109		
Kursus ini merangkumi metabolisme asid amino, protein, nukleotida dan asid nukleik di dalam sel. Integrasi metabolisme protein dan asid nukleik serta pengawalannya dibincangkan. Penganalisaan tindak balas metabolisme protein dan asid nukleik turut dijalankan		
<i>This course encompasses metabolism of amino acids, proteins, nucleotides and nucleic acids in cells. Metabolisms of proteins and nucleic acids and their regulations are discussed. Analyses of protein and nucleic acid metabolic reactions are conducted.</i>		

BCH3111	Metabolisme Lipid dan Membran / Lipid Metabolism and Membranes	4(3+1)
Prasyarat: BCH3108		
Kursus ini merangkumi metabolisme lipid dan membran di dalam sel. Pengawalan metabolisme lipid dan membran dibincangkan. Penganalisaan hasil tindak balas metabolisme lipid dan membran turut dijalankan.		
<i>This course encompasses cellular metabolisms of lipid and membrane. Regulation of lipid and membrane metabolisms are discussed. Analysis of lipid and membrane metabolic reactions are conducted.</i>		

BCH3204	Analisis Biomolekul / Analysis of Biomolecules	3(2+1)
Prasyarat: BCH3004 atau BCH3107		
Kursus ini merangkumi teknik penganalisaan karbohidrat, lipid, protein dan asid nukleik. Analisis sifat fizikal dan kimia, berat molekul dan kehomogenan biomolekul dibincangkan. Penganalisaan biomolekul turut dijalankan		
<i>This course encompasses analytical techniques of water content, acidity, carbohydrates, lipids, proteins and nucleic acids. Analysis of physical and chemical properties, molecular weight and homogeneity of biomolecules is discussed. Analyses of biomolecules are conducted.</i>		

BCH3301	Prinsip Metabolomik / Principles of Metabolomics	3(3+0)
Prasyarat: BCH3004 atau BCH3111		
Kursus ini merangkumi prinsip analisis komponen metabolit dalam tumbuhan, haiwan, mikroba, alam sekitar dan makanan. Teori penggunaan peralatan terkini dalam metabolomik ditekankan. Penganalisaan data metabolomik menggunakan pangkalan data juga dijalankan.		

This course encompasses analytical techniques of metabolite components in plants, animals, microbes, environment and foods. Theory of using advanced instruments in metabolomics is emphasized. Analyses of metabolomics data using related software and database are also conducted.

BCH4101	Biokimia Hormon / Biochemistry of Hormones	3(3+0)
Prasyarat: BCH3004 atau BCH3111		
Kursus ini merangkumi pengkelasan, sifat kimia dan penghasilan hormon. Fungsi hormon dalam metabolisme sel, koordinasi sel, interaksi antara molekul dan kecacatan patofisiologi turut dibincangkan.		
<i>This course encompasses the classification, chemical properties and production of hormones. The function of hormones in cell metabolism, cell coordination, interactions among molecules and pathophysiological disorders are also discussed.</i>		

BCH4301	Biokimia Tumbuhan / Plant Biochemistry	3(3+0)
Prasyarat: BCH3108 atau BCH3004		
Kursus ini merangkumi pengkelasan metabolit primer dan sekunder serta kepelbagaian sistem metabolik dalam tumbuhan. Integrasi laluan metabolik dalam tumbuhan dan pengawalannya dibincangkan		
<i>This course encompasses classification of primary and secondary metabolites as well as various metabolic systems in plant. Metabolic integration in plants and its regulations are discussed.</i>		

BCH4302	Manipulasi Genetik Tumbuhan / Plant Genetic Manipulation	3(2+1)
Prasyarat: BCH4301		
Kursus ini merangkumi pelbagai teknik manipulasi gen tumbuhan. Kesan penggunaan teknologi manipulasi gen tumbuhan kepada ekonomi, alam sekitar dan sosio-budaya dibincangkan. Manipulasi gen tumbuhan dijalankan.		
<i>This course encompasses plant gene manipulation techniques. Implication of plant gene manipulation technologies towards economy, environment and sociocultural are discussed. Plant gene manipulation is conducted.</i>		

BCH4303	Aplikasi Biokimia dalam Industri / Industrial Applications of Biochemistry	3(3+0)
Prasyarat: BCH3004 atau BCH3108		
Kursus ini merangkumi konsep biokimia dalam penghasilan produk. Aplikasi biokimia dalam pelbagai industri seperti farmaseutika, kimia, makanan dan diagnostik dibincangkan. Kepentingan paten dan aspek keselamatan juga diuraikan.		
<i>This course encompasses biochemistry concepts in product manufacturing. Application of biochemistry in various industries such as pharmaceuticals, chemicals, food and diagnostic are discussed. The importance of patent and safety aspects are also described.</i>		

BCH4304	Biokimia Makanan / Food Biochemistry	3(3+0)
Prasyarat : BCH3108 atau BCH3004		
Kursus ini merangkumi ciri fisiko-kimia makanan dan komponen utamanya. Aktiviti mikrob, kaedah pengawetan makanan serta penggunaan enzim dalam teknologi makanan dibincangkan. Kesan tindak balas biokimia terhadap perubahan luaran serta nilai nutrisi makanan turut dibincangkan.		
<i>This course encompasses physico-chemical properties of foods and their main components. Microbial activity, food preservation methods and the use of enzymes in food technology are discussed. Effects of biochemical reactions on appearance and the nutritional value of food are also discussed.</i>		

BCH4305	Biokimia Pemakanan / Nutritional Biochemistry	3(3+0)
Prasyarat: BCH3108 atau BCH3004		
Kursus ini merangkumi peranan nutrien dalam fisiologi manusia dan haiwan. Keperluan pemakanan dan kesan ketidakseimbangan nutrien ke atas kesihatan manusia dan haiwan dibincangkan. Kaedah penilaian pemakanan turut dibincangkan.		
<i>This course encompasses the role of nutrients in human and animal physiology. Nutritional requirements and effects of nutrient imbalance towards human and animal health are discussed. Dietary assessment methods are also discussed.</i>		

BCH4306	Biokimia Tisu Haiwan / Biochemistry of Animal Tissues	3(3+0)
Prasyarat: BCH3108 atau BCH3004		
Kursus ini merangkumi proses biokimia di dalam pelbagai jenis tisu haiwan. Integrasi metabolisme di antara tisu dibincangkan. Patofisiologi yang berkaitan dengan biokimia tisu haiwan turut dibincangkan.		
<i>This course encompasses biochemical processes in various tissues of animals. Metabolic integrations between tissues are discussed. Pathophysiological related to animal tissue biochemistry are also discussed.</i>		

BCH4307	Biokimia Alam Sekitar/ Environmental Biochemistry	3(3+0)
Prasyarat: BCH3108 atau BCH3004		
Kursus ini merangkumi proses penguraian jisim organik, rawatan air buangan dan penyesuaian mekanisme molekul terhadap persekitaran lampau. Kesan bioakumulasi, ketoksikan, rintangan pencemar persekitaran dan kepatogenan terhadap metabolisme sel, alam sekitar, ekonomi dan kesihatan dibincangkan		
<i>This course encompasses the process of decomposition of organic matter, waste water treatment and the molecular mechanisms of adaptation to extreme environments. Effects of bioaccumulation, development of resistance to environmental contaminants and pathogenicity towards cellular metabolism, environment, economy and health are discussed.</i>		

BCH4309	Biokimia Pengangkutan dan Disposisi Ubatan / Biochemistry of Drug Transportation and Disposition	3(3+0)
Prasyarat : BCH3004 atau BCH3111		
Kursus ini merangkumi peranan yang dimainkan oleh pengangkut membran dalam pelupusan dan tindakan ubatan. Konsep utama mengenai interaksi pengangkut dengan metabolisme enzim serta interaksi antara ubatan dan makanan turut diuraikan.		
<i>This course encompasses the role played by membrane transporters in drug disposition and action. The main concepts of transport interactions with drug metabolizing enzymes as well as interactions between drugs and food are also elaborated.</i>		

BCH4310	Biokimia Toksikologi Haiwan / Biochemistry of Animal Toxicology	3(3+0)
Prasyarat: BCH3004 atau BCH3111		
Kursus ini merangkumi prinsip asas penyerapan, pengedaran, metabolisme dan perkumuhan toksikan. Farmakokinetik, mekanisme dan prinsip toksikologi dibincangkan. Farmakogenetik yang berkaitan dengan pembangunan ubatan dan penilaian keselamatan juga ditekankan.		
<i>This course encompasses the fundamental principles of absorption, distribution, metabolism and excretion of toxicants. The pharmacokinetics, mechanisms and toxicology principles are discussed. The pharmacogenetics related to drug development and safety evaluation is also emphasized.</i>		

BCH4311	Sains Data dalam Biokimia / Data Science in Biochemistry	3(3+0)
Prasyarat: BTC3005 dan BCH3004 atau BCH3111		
Kursus ini mengandungi pembelajaran berkenaan takrifan asas dan penggunaan data dalam biokimia. Kefahaman dan kemahiran bioanalitik, pengelasan dan penggunaan data raya ditekankan. Komunikasi pengaplikasian sains data dalam biokimia turut dibincangkan.		
<i>This course encompasses the learning of basic definition and application of data in biochemistry. Understanding and skills of bioanalysis, classification and use of big data are emphasized. Communication in data science application in biochemistry is also discussed.</i>		
BCH4312	Biokimia Isyarat Tumbuhan / Biochemistry of Plant Signaling	3(3+0)
Prasyarat: BCH4301		
Kursus ini merangkumi transduksi isyarat dan pengawalannya dalam tumbuhan. Penglibatan fitohormon, pengutus isyarat kedua dan pengatur protein dalam setiap transduksi isyarat dibincangkan. Cakap silang antara laluan transduksi isyarat dalam pelbagai keadaan juga turut diuraikan.		
<i>This course encompasses signal transduction pathway and its regulation. Involvement of phytohormones, second messengers and protein regulators in each signal transduction pathway is discussed. Crosstalk between signal transduction pathways in various conditions is also explained.</i>		
BCH4904	Pembelajaran Perkhidmatan dalam Biokimia/ Service Learning in Biochemistry	1(0+1)
Prasyarat: BCH3109 atau BCH3110 atau BCH3111		
Kursus ini merangkumi aktiviti di antara pelajar dan komuniti bagi meningkatkan kesedaran dan kefahaman komuniti tentang kepentingan biokimia dalam kehidupan seharian. Perkhidmatan komuniti berdasarkan aplikasi pengetahuan biokimia ditekankan. Kerjasama bagi membina pengetahuan sains di kalangan masyarakat turut ditekankan.		
<i>This course encompasses activities between students and communities to increase awareness and understanding of the importance of biochemistry in daily lives among communities. Community services through application of biochemistry knowledge are emphasized. Cooperation towards enhancing scientific knowledge among society is also emphasized.</i>		
BCH4905	Latihan Industri / Industrial Training	12(0+12)
Prasyarat: BCH3109 atau BCH3110 atau BCH3111		
Kursus ini mendedahkan pelajar kepada persekitaran kerja yang sebenar di industri/organisasi selama 24 minggu. Latihan merangkumi aplikasi aspek teori dan praktikal yang telah dipelajari dengan amalan semasa di tempat kerja. Kemahiran menyelesaikan masalah dan komunikasi turut diberi penekanan.		
<i>This course introduces students to the real working environment in industries/ organizations. Training includes application of the theoretical and practical aspects that have been studied with current practices in the workplace. Problem solving and communication skills are also emphasized.</i>		

BCH4906	Topik Khas dalam Biokimia / Current Topics in Biochemistry	1(1+0)
Prasyarat: BCH3109 atau BCH3110 atau BCH3111		
Kursus ini memberi tumpuan kepada masalah, isu dan trend terpilih berkaitan bidang biokimia. Penerokaan isu penting dan hala tuju baharu dibuat melalui analisis isu kritikal dan permasalahan bagi mencadangkan penyelesaian. Peluang perniagaan dalam pengkomersilan produk berkaitan bidang biokimia dibincangkan.		
<i>This course focuses on selected problems, issues and trends related to the biochemistry field. Exploration of key issues and new direction is conducted through analysis of critical issues and problems to recommend solutions. Business opportunities in commercialization of products related to the biochemistry field is discussed.</i>		

BCH4959	Disertasi Bachelo / Bachelor Dissertation	6(0+6)
Prasyarat: BCH3109 atau BCH3110 atau BCH3111		
Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui reka bentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.		
<i>This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.</i>		

2. Jabatan Mikrobiologi / Department of Microbiology

BMY3001	Mikrobiologi / Microbiology	4(4+0)
Prasyarat: Tiada		
Kursus ini merangkumi beberapa aspek mikrobiologi asas termasuk organisasi dan struktur mikrob, mikroskopi, sistematik dan ekologi mikrob. Pengelasan dan pengenalpastian mikroorganisma juga dibincangkan.		
<i>This course encompasses several aspects of basic microbiology including the microbial organization and structure, microscopy, microbial systematics and ecology. Classification and identification of microorganisms are also described.</i>		
BMY3204	Teknik Asas Mikrobiologi / Basic Microbiology Techniques	3(1+2)
Prasyarat: Tiada		
Kursus ini merangkumi teknik asas mikrobiologi dalam mengendalikan kultur mikroorganisma. Penggunaan mikroskop dan pelbagai jenis kaedah pewarnaan ditekankan. Keperluan fizikal dan kimia dalam pertumbuhan mikroorganisma, serta enumerasi mikroorganisma juga dibincangkan.		
<i>This course encompasses basic microbiology techniques in handling microbial cultures. The use of microscope and various staining techniques are emphasized. Physical and chemical requirements in microbial growth and enumeration techniques are discussed.</i>		
BMY3205	Teknik Pencirian Mikroorganisma / Techniques in Microbial Characterization	3(1+2)
Prasyarat: BMY3204		
Kursus ini merangkumi analisis biokimia untuk pengenalpastian mikroorganisma dari pelbagai sumber. Pelbagai teknik dalam pengenalpastian mikrob dan pengiraan virus dijalankan. Kesan faktor fizikal dan kimia terhadap pertumbuhan mikroorganisma juga dikaji.		
<i>This course encompasses biochemical tests for the identification of microorganisms from various sources. Techniques in microbial identification and virus enumeration are conducted. Effects of physical and chemical factors on microbial growth are also studied.</i>		
BMY3206	Teknik Mikrobiologi Lanjutan / Advanced Microbiological Techniques	4(1+3)
Prasyarat: BMY3204		
Kursus ini merangkumi beberapa teknik lanjutan dalam mikrobiologi yang menekankan hubungkait mikroorganisma seperti kulat, bakteria anaerobik, dan virus dengan persekitaran. Pelbagai teknik pertumbuhan dan kuantifikasi mikroorganisma ditekankan.		
<i>This course encompasses several techniques in microbiology that emphasises the connection of microorganisms such as fungi, anaerobic bacteria, and virus with environment. Several cultural and quantification techniques of microorganism are highlighted.</i>		

BMY4202	Teknik Khas dalam Mikrobiologi / Specialise Techniques in Microbiology	4(1+3)
Prasyarat: BMY3206		
Kursus ini merangkumi teknik khas yang menekankan peranan mikroorganisma dalam kehidupan harian. Kegunaan prinsip imunologi dalam teknik diagnostik klinikal dan kaedah bioinformatik dalam menganalisis fungsi mikroorganisma dibincangkan. Teknik penanaman cendawan, pengkulturan sel haiwan dan penghasilan makanan melalui fermentasi juga ditekankan.		
<i>This course encompasses special techniques that emphasize the role of microorganisms in daily life. The application of immunological principles in clinical diagnostic techniques and bioinformatics methods in analyzing the function of microorganisms are discussed. Techniques of mushroom cultivation, animal cell culturing and food production through fermentation are also emphasized.</i>		
BMY4302	Virologi / Virology	3(3+0)
Prasyarat: BMY3001		
Kursus ini merangkumi pencirian virus termasuk prion dan viroid. Pengkelasan dan taksonomi virus diterangkan. Tindak balas virus-sel perumah, epidemiologi dan kepatogenan dihuraikan. Etika dan biokeselamatan juga dibincangkan.		
<i>This course encompasses characterization of viruses including prions and viroids. Classification and taxonomy of viruses are described. Virus-host cell interactions, epidemiology and pathogenicity are described. Ethics and biosafety will also be discussed.</i>		
BMY4303	Mikologi / Micology	3(3+0)
Prasyarat: BMY3001		
Kursus ini merangkumi pengkelasan, struktur, pertumbuhan dan pembiakan, genetik dan interaksi fungus sebagai saprofit dan parasit. Peranan fungus dalam pencegahan dan pengawalan pertumbuhan mikrob dibincangkan.		
<i>This course encompasses classification, structure, growth and reproduction, genetics and interactions of fungi as saprophytes and parasites. The roles of fungi in the prevention and control of microbial growth are discussed.</i>		
BMY4304	Imunologi / Immunology	3(3+0)
Prasyarat: BMY3001		
Kursus ini merangkumi pengkelasan, struktur, pertumbuhan dan pembiakan, genetik dan interaksi fungus sebagai saprofit dan parasit. Peranan fungus dalam pencegahan dan pengawalan pertumbuhan mikrob dibincangkan.		
<i>This course encompasses several concepts in immunology. Immune system regulations and their roles in diseases are discussed. Research and current developments in immunology are described.</i>		
BMY4305	Ekologi Mikrob / Microbial Ecology	3(3+0)
Prasyarat: BMY3001		
Kursus ini merangkumi pelbagai aspek ekologi mikrob yang melibatkan hubungan dan interaksi sesuatu mikroorganisma dengan persekitarannya. Kaedah dan teknologi yang digunakan untuk mengkaji mikrob dan aktiviti dibincangkan. Kepentingan ekonomi dan biodiversiti populasi mikrob juga dihuraikan.		
<i>This course encompasses various aspects of microbial ecology which involve the interaction between microorganisms with their environment. Methods and technology used to study microbes and their activities are discussed. The economic importance and biodiversity of microbial population are also elaborated.</i>		

BMY4306	Mikroorganisma dan Penyakit / <i>Microorganism and Disease</i>	3(3+0)
Prasyarat: BMY3001		
Kursus ini menekankan kepentingan mikrobiologi dalam perubatan. Interaksi antara bakteria, virus, dan kulat patogenik dengan hos dibincangkan. Diagnosis, rawatan dan pencegahan jangkitan diuraikan.		
<i>This course emphasizes the importance of microbiology in medicine. Interactions between bacterial, viral, and fungal pathogens with their hosts are discussed. Diagnosis, treatment and prevention of infections are described.</i>		

BMY4307	Fisiologi Kulat / <i>Fungal Physiology</i>	3(3+0)
Prasyarat: BMY4303		
Kursus ini merangkumi komposisi kimia dan struktur molekul sel kulat. Keperluan fizikal dan kimia untuk pertumbuhan kulat, metabolisme primer dan sekunder serta genetik dan perkembangan spora turut dibincangkan. Kerintangan dan mekanisme pertahanan terhadap fungisid diuraikan.		
<i>This course encompasses the chemical composition and molecular structures of fungal cells. The physical and chemical requirements for fungal growth, primary and secondary metabolisms, genetics and spore development are discussed. Resistance and defensive mechanisms against fungicides are described.</i>		

BMY4309	Mikrobiologi Makanan Gunaan / <i>Applied Food Microbiology</i>	3(3+0)
Prasyarat: BMY3001		
Kursus ini merangkumi isu semasa mikrobiologi makanan gunaan termasuk kemunculan patogen bawaan makanan baharu, kaedah pantas pengenalpastian dan pencirian mikroorganisma makanan. Kesan komposisi dan kaedah pengawetan makanan diuraikan. Ciri fungsian mikroorganisma makanan terpilih dan aplikasi dalam industri dibincangkan.		
<i>This course encompasses current issues in applied food microbiology which include new and emerging food-borne pathogens, rapid identification and characterization of food microorganisms. The effects of food compositions and preservation techniques are described. Functional properties of selected food microorganisms and their industrial applications are discussed.</i>		

BMY4310	Genetik Mikrob / <i>Microbial Genetics</i>	4(3+1)
Prasyarat: BMY3001		
Kursus ini merangkumi pelbagai aspek genetik mikrob seperti replikasi DNA, pengawalan pengekspresan gen, mekanisme pemindahan DNA dan rekombinasi genetik. Unsur dan mekanisme transposisi serta teknik molekul dalam genetik mikrob dan gunaan dibincangkan.		
<i>This course encompasses various aspects of microbial genetics such as DNA replication, control of gene expression, mechanisms of DNA transfer and genetic recombination. Elements and mechanisms of transposition as well as molecular techniques in microbial and applied genetics are discussed.</i>		

BMY4311	Bakteriologi / <i>Bacteriology</i>	4(4+0)
Prasyarat: BMY3001		
Kursus ini merangkumi pengelasan, pencirian, ekologi dan aktiviti pelbagai jenis bakteria. Struktur dan aktiviti metabolik bakteria, metabolisme berpusat dan tindak balas penjanaan tenaga bakteria diuraikan. Biogenesis dan fungsi makromolekul serta integrasi proses metabolik dibincangkan. Kegunaan bakteria dalam industri dan pertanian turut diberi penekanan.		
<i>This course encompasses classification, characterisation, ecology and activities of various types of bacteria. The bacterial structure, metabolic activities, central metabolism and energy production reactions are described. Macromolecular biogenesis and functions as well as the integration of bacterial metabolic processes are discussed. The use of bacteria in the industries and agriculture are also emphasized.</i>		

BMY4902	Latihan Industri / Industrial Training	12(0+12)
Prasyarat: Dengan kebenaran jabatan		
Kursus ini mendedahkan pelajar kepada persekitaran kerja yang sebenar di industri/organisasi. Latihan industri ini dijalankan selama 24 minggu. Ianya merangkumi aplikasi aspek teori dan praktikal yang telah dipelajari dengan amalan semasa di tempat kerja. Kemahiran menyelesaikan masalah dan komunikasi turut diberi penekanan.		
<i>This course introduces students to real working environment in industries/organizations. This industrial training is conducted for 24 weeks. Training includes application of the theoretical and practical aspects that have been studied with current practices in the workplace. Problem solving and communication skills are also emphasized.</i>		
BMY4904	Pembelajaran Perkhidmatan dalam Mikrobiologi / Service Learning in Microbiology	1(0+1)
Prasyarat: BMY3206		
Kursus ini merangkumi aktiviti bagi meningkatkan kesedaran dan kefahaman komuniti tentang kepentingan mikroorganisma dalam kehidupan harian. Pengetahuan mikrobiologi diaplikasikan melalui aktiviti kerjasama dengan masyarakat luar.		
<i>This course encompasses activities to increase awareness and understanding of the community on the importance of microbiology in daily lives. Knowledge in microbiology is applied through joined activities with the community.</i>		
BMY4959	Disertasi Bacelor / Bachelor Dissertation	6(0+6)
Prasyarat: Tiada		
Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui rekabentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.		
<i>This course covers the preparation of proposals, implementation and scientific writing of research projects. Scientific approaches to generate data systematically through appropriate design, data collection and analysis are emphasized.</i>		
BMY4991	Seminar / Seminar	1(0+1)
Prasyarat: BMY3206		
Kursus ini merangkumi penyediaan dan penyampaian seminar secara berkesan mengenai projek penyelidikan dalam bidang mikrobiologi. Sorotan literatur, penyusunan dan pembentangan maklumat mikrobiologi dilakukan dalam bentuk seminar.		
<i>This course encompasses the effective preparation and delivery of seminars on research project in microbiology. Literature review, organization and presentation of microbiology information is done in the form of seminars.</i>		
BMY4993	Komunikasi Sains / Science Communication	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi kemahiran dan teknik komunikasi untuk menyampaikan maklumat sains secara lisan dan bertulis kepada khalayak sasaran. Penyampaian maklumat sains melalui seni juga diterapkan. Pengurusan maklumat dan pengenalanpastian kredibiliti sumber maklumat juga ditekankan.		
<i>This course covers communication skills and techniques for conveying scientific information orally and in written presentations to the targeted audience. Delivery of science through arts is also embedded. Information management and identification of information source credibility are also emphasised.</i>		

3. Jabatan Teknologi Bioproses / Department of Bioprocess Technology

BTC3000	Kimia Biofizikal / Biophysical Chemistry	3(2+1)
Prasyarat: Tiada		
Kursus ini merangkumi prinsip kimia biofizikal yang berkaitan dengan sains biologi. Pengetahuan asas kimia biofizikal berkaitan dengan biomakromolekul dan interaksi fizikalnya diuraikan. Aplikasi teknik kuantitatif dalam pengukuran ciri fisiko-kimia biomolekul turut dibincangkan.		
<i>This course encompasses principles of biophysical chemistry related to life science. Fundamental knowledge of biophysical chemistry related to biomacromolecules and their physical interaction is elaborated. The application of quantitative techniques in the measurement of physico-chemical properties of biomolecules is also discussed.</i>		
BTC3001	Pengenalan Kepada Bioteknologi / Introduction to Biotechnology	2(2+0)
Prasyarat: Tiada		
Kursus ini merangkumi pengenalan asas kepada pelbagai bidang bioteknologi. Kemajuan dan isu terkini dalam bidang bioteknologi turut dibincangkan secara pembentangan seminar.		
<i>This course encompasses the introduction on various biotechnology fields. Recent advances and issues in biotechnology will also be discussed through seminar presentation.</i>		
BTC3002	Komersialisasi dan Isu Semasa Bioteknologi / Commercialisation and Current Issues In Biotechnology	2(2+0)
Prasyarat: Tiada		
Kursus ini merangkumi konsep dan langkah dalam pengkomersilan produk bioteknologi. Topik terkini dan kemajuan yang berkaitan dengan pengkomersilan produk bioteknologi turut dibincangkan.		
<i>This course encompasses concept and steps in the commercialization of biotechnology products. Current topics and latest development related to commercialization of biotechnology products are also discussed.</i>		
BTC3003	Instrumentasi dalam Penyelidikan Bioteknologi / Instrumentation in Biotechnology Research	3(2+1)
Prasyarat: Tiada		
Kursus ini merangkumi prinsip operasi dan cara penggunaan instrumen saintifik dalam penyelidikan bioteknologi. Kaedah interpretasi data penyelidikan yang terhasil dari kajian saintifik terpilih yang dijalankan juga dibincangkan.		
<i>This course encompasses the principles of scientific instrument operation in biotechnology research. Method for research data interpretation resulting from the selected scientific experiment is also discussed</i>		
BTC3004	Penulisan Saintifik dalam Bioteknologi / Scientific Writing in Biotechnology	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi kaedah untuk mencari dan menilai maklumat bidang bioteknologi daripada pelbagai sumber untuk penulisan saintifik. Beberapa manuskrip terpilih juga dibincangkan.		
<i>This course encompasses methods to search and evaluate information related to the biotechnology field from various sources for scientific writing. Several selected manuscripts are also discussed.</i>		

BTC3005	Statistik dan Biometri dalam Bioteknologi / <i>Statistics and Biometrics in Biotechnology</i>	3(2+1)
Prasyarat: Tiada		
Kursus ini merangkumi kaedah penganalisan data dan mereka bentuk eksperimen bioteknologi. Statistik diskriptif dan inferensi, analisis univariat dan multivariat, ujian keertian dan pengurusan ralat ke atas set data, dan perlombongan data dari keputusan eksperimen multi dimensi juga dibincangkan.		
<i>This course encompasses methods of data analysis and biotechnology experimental design. Descriptive and inferential statistics, univariate and multivariate analysis, significance tests and management of error on dataset, and data mining from multi-dimensional experimental results are also discussed.</i>		

BTC3101	Teknologi Enzim / <i>Enzyme Technology</i>	4(3+1)
Prasyarat: BCH3004 atau BCH3108		
Kursus ini merangkumi aspek penghasilan, pengekstrakan dan penulenan enzim intrasel dan ekstrasel. Penggunaan enzim dalam pelbagai bidang turut dibincangkan.		
<i>This course encompasses various aspects of production, extraction and purification of intracellular and extracellular enzymes. The use of enzyme in various fields is also discussed.</i>		

BTC3201	Teknologi Fermentasi / <i>Fermentation Technology</i>	4(3+1)
Prasyarat: BMY3001		
Kursus ini merangkumi pelbagai aspek penting di dalam teknologi fermentasi yang melibatkan langkah sebelum dan semasa proses fermentasi serta analisis selepas proses fermentasi. Aplikasi teknologi fermentasi di dalam industri dibincangkan.		
<i>Kursus ini merangkumi pelbagai aspek penting di dalam teknologi fermentasi yang melibatkan langkah sebelum dan semasa proses fermentasi serta analisis selepas proses fermentasi. Aplikasi teknologi fermentasi di dalam industri dibincangkan.</i>		

BTC3301	Kejuruteraan Bioproses / <i>Bioprocess Engineering</i>	4(4+0)
Prasyarat: BTC3201		
Kursus ini merangkumi aspek yang berkaitan kejuruteraan bioproses seperti asas pengiraan kejuruteraan dan pengoptimuman proses. Pengenalan kepada peralatan industri dan aspek kejuruteraan di dalam operasi bioreaktor untuk kultur sel dan tindakbalas enzim turut dibincangkan.		
<i>This course encompasses aspects related to bioprocess engineering such as fundamental engineering calculation and process optimisation. Overview on the industrial equipment and the engineering aspects in the bioreactor operation for cell cultures and enzyme reaction are discussed.</i>		

BTC3302	Biopemisahan dan Penulenan / <i>Bioseparation and Purification</i>	4(3+1)
Prasyarat: BTC3201		
Kursus ini merangkumi pemprosesan hiliran untuk pemulihan semula dan penulenan produk bioteknologi. Pelbagai teknik biopemisahan dan penulenan dibincangkan melalui carta alir, analisisimbangan jisim dan kos.		
<i>This course encompasses the downstream processing for the recovery and purification of biotechnological products. Various bioseparation and purification techniques are discussed using flow chart, mass balance analysis and costing.</i>		

BTC3305	Reka Bentuk Bioproses dan Biopenghasilan / Bioprocessing and Biomanufacturing Design	3(2+1)
Prasyarat: BTC3301		
Kursus ini merangkumi aspek rekaan dan penggabungan pelbagai unit operasi yang terlibat di dalam penghasilan, biopemisahan, penulenan dan formulasi produk bioteknologi. Sintesis dan analisis proses, serta kajian kes yang berkaitan dengan reka bentuk bioproses dibincangkan. Pembelajaran berasaskan projek juga dilibatkan.		
<i>This course encompasses aspects of designing and consolidating various unit operations involved in the production, bioseparation, purification and formulation of biotechnological products. Process synthesis and analysis as well as case studies related to bioprocess are discussed. Project based learning is also included.</i>		

BTC3306	Sistem Bioreaktor / Bioreactor System	3(3+0)
Prasyarat: BTC3201 dan BTC3301		
Kursus ini merangkumi pelbagai aspek sistem bioreaktor yang digunakan dalam proses penghasilan produk bioteknologi menggunakan mikroorganisma, enzim, sel haiwan dan tumbuhan. Keperluan spesifik terhadap reka bentuk dan pendekatan peningkatan skala bioreaktor untuk sistem biologi yang berbeza dibincangkan.		
<i>This course encompasses various aspects of bioreactor system used in the production of biotechnological products employing microorganisms, enzymes, animal and plant cells. Specific requirement for the design and scaling-up approaches of bioreactor for different biological systems are discussed.</i>		

BTC3401	Pengurusan dan Penggunaan Sisa / Waste Management and Utilization	2(2+0)
Prasyarat: Tiada		
Kursus ini merangkumi perincian sumber, jenis, ciri dan komposisi sisa, rawatan dan pengendalian sisa. Kajian kes termasuk perbandingan kaedah pengurusan sisa di Malaysia dan di negara lain dibincangkan.		
<i>This course encompasses details on the sources, types, characteristics and composition of wastes, waste handling and management. Case studies includes comparison of waste management methods in Malaysia and other countries are discussed.</i>		

BTC3501	Konsep Biokeusahawanan / Bioentrepreneurship Concept	3(2+1)
Prasyarat: MGM3180		
Kursus ini merangkumi konsep biokeusahawanan yang melibatkan elemen seperti pemilihan produk dan perkhidmatan bioteknologi, jenis perniagaan dan analisis pasaran. Aktiviti kajian pasaran bagi pengkomersilan produk dan perkhidmatan bioteknologi ditekankan.		
<i>This course encompasses bioentrepreneurship concept involving elements such as selection of biotechnological products and services, types of business and market analysis. Market survey activity for commercialization of biotechnological products and services is emphasized.</i>		

BTC3502	Perancangan Biokeusahawanan / Bioentrepreneurship Planning	3(2+1)
Prasyarat: BTC3501		
Kursus ini merangkumi penyediaan rancangan perniagaan bagi produk atau perkhidmatan bioteknologi. Elemen inovasi dan teknik kreatif dalam membina konsep produk turut dibincangkan.		
<i>This course encompasses preparation of a business plan for biotechnological products or services. Innovation element and creative techniques in developing product concept are also discussed.</i>		

BTC4001	Biokeselamatan Dan Bioetika / Biosafety And Bioethics	2(2+0)
Prasyarat: Tiada		
Kursus ini merangkumi isu biokeselamatan dan bioetika dalam bioteknologi, termasuk tahap biokeselamatan dalam reka bentuk makmal. Isu berkaitan keselamatan dan etika, serta cara pengendalian isu dan pelaksanaan keputusan yang rasional turut dibincangkan.		
<i>This course covers biosafety and bioethics issues in biotechnology including safety level in laboratory design. Issues on safety and ethics, including the handling of biological issues and rational decision-making are also discussed.</i>		
BTC4002	Bioteknologi dalam Bioekonomi / Biotechnology in Bioeconomy	2(2+0)
Prasyarat: BTC3002		
Kursus ini merangkumi konsep dan isu berkaitan aplikasi bioteknologi dalam penjaan bioekonomi. Kepentingan teknologi berasaskan biosumber serta sumbangan bioteknologi kepada penjaan ekonomi secara mampan dibincangkan. Kebaikan dan cabaran bioekonomi peringkat global juga diuraikan.		
<i>This course covers concept and issues related to biotechnology in bioeconomy. The importance of bioresources and contribution of biotechnology towards sustainable generation of economy are discussed. The advantages and challenges of bioeconomy globally are also elaborated.</i>		
BTC4003	Transisi ke Arah Bioekonomi Kitaran / Transition towards Circular Bioeconomy	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi pelbagai faktor yang merangsang kemajuan negara dalam bioekonomi kitaran ke arah pembangunan lestari. Pelbagai strategi dan polisi berkaitan bioekonomi kitaran serta trend permintaan dan penawaran yang didorong oleh perubahan dalam persekitaran pasaran turut dibincangkan.		
<i>This course encompasses various factors that stimulate the country progress in circular bioeconomy towards sustainable development. Various strategies and policies related to circular bioeconomy as well as the trends in demand and supply driven by the changes in market environment are also discussed.</i>		
BTC4104	Bioteknologi Makanan / Food Biotechnology	3(2+1)
Prasyarat: BTC3101		
Kursus ini merangkumi penggunaan biokatalis dalam industri makanan dan ramuan makanan. Pengubahsuaian makanan, makanan transgenik dan isu semasa serta undang-undang berkaitan bioteknologi makanan turut dibincangkan.		
<i>This course involves the applications of biocatalysts in the food and food ingredient industries. Modification of foods, transgenic foods and current issues as well as laws related to food biotechnology are discussed.</i>		
BTC4105	Bioteknologi Makanan Lanjutan / Advanced Food Biotechnology	2(2+0)
Prasyarat: BTC4104		
Kursus ini merangkumi peranan dan sumbangan bioteknologi makanan terhadap gaya hidup masyarakat, keselamatan dan kesejahteraan sosial. Perkembangan terkini dalam bidang bioteknologi makanan dibincangkan.		
<i>This course encompasses the roles and contributions of food biotechnology towards people's lifestyle, security and social well-being. Latest development in food biotechnology is discussed.</i>		

BTC4106	Teknologi Enzim Lanjutan / Advanced Enzyme Technology	3(2+1)
Prasyarat: BTC3101		
Kursus ini merangkumi aspek penyekatgerakan enzim dan sel dan aplikasi enzim tersekatgerak dalam industri. Penggunaan enzim dan sel tersekatgerak dalam biosensor dan media bukan konvensional dibincangkan.		
<i>This course encompasses immobilization of enzymes and cells and application of immobilized enzyme in the industry. The use of immobilized enzymes and cells in biosensor and non-conventional media are discussed.</i>		
BTC4205	Mikrobiologi Industri / Industrial Microbiology	3(2+1)
Prasyarat: BMY3001		
Kursus ini merangkumi kaedah peningkatan hasil produk daripada mikroorganisma yang digunakan dalam industri. Pelbagai aplikasi dalam penghasilan produk mikrobiologi industri turut dibincangkan.		
<i>This course encompasses methods of increasing product yield from microorganisms used in industry. A variety of applications in the production of industrial microbiology products are discussed.</i>		
BTC4305	Pemodelan Dan Pengoptimuman Bioproses / Bioprocess Modelling and Optimization	3(2+1)
Prasyarat: BTC3201 dan BTC3301		
Kursus ini merangkumi penggunaan teknik pengoptimuman dan simulasi dalam proses biologi. Program simulasi komputer digunakan untuk pemodelan persamaan matematik, analisis statistik dan pengoptimuman parameter bioproses.		
<i>This course encompasses application of optimisation and simulation technique in biological process. Computer simulation programs are used for modeling of mathematical equations, statistical analysis and bioprocess parameters optimisation.</i>		
BTC4306	Formulasi Produk Akhir Bioteknologi / Formulation of Biotechnological End Product	3(2+1)
Prasyarat: BTC3002		
Kursus ini merangkumi aspek penting dalam formulasi pelbagai bentuk fizikal produk siap bioteknologi yang melibatkan bahan dan proses tertentu. Kaedah pembungkusan, penyimpanan dan penentuan jangka hayat produk turut dibincangkan.		
<i>This course encompasses the important aspects in formulation of different physical forms of biotechnological finished products that involves certain materials and processes. Techniques in product packaging, storage and determination of shelf-life are also discussed.</i>		
BTC4307	Teknologi Pengekstrakan Sebatian Bioaktif / Bioactive Compounds Extraction Technology	3(2+1)
Prasyarat: BTC3001		
Kursus ini merangkumi prinsip dan aspek asas pengekstrakan pelbagai sumber sebatian bioaktif. Isu berkaitan teknik pengekstrakan serta trend dan aplikasi sebatian bioaktif yang terlibat dibincangkan. Proses pasca pengekstrakan dan keperluan berkaitan kualiti dan isu keselamatan turut diberi penekanan.		
<i>This course covers basic principles and aspects of extracting various sources of bioactives compounds. Issues related to extraction techniques as well as trends and applications involved are discussed. Post-extraction processes and requirements related to quality and safety issues are also emphasized.</i>		

BTC4406	Bioremediasi / Bioremediation	3(3+0)
Prasyarat: BTC3401		
Kursus ini merangkumi kaedah rawatan dan bioremediasi ke atas sisa toksik terutama terhadap tanah dan air bawah tanah yang tercemar. Potensi penggunaan mikroorganisma untuk pelbagai rawatan sisa toksik dan berbahaya dibincangkan. Proses yang diguna pakai dalam industri dirujuk untuk menjelaskan kaedah dan konsep bioremediasi.		
<i>This course encompasses treatment methods and bioremediation of toxic waste especially in polluted soil and groundwater. The potential use of microorganisms in various treatments of toxic and hazardous wastes are discussed. Industrial processes are referred in order to clarify the method and concept of bioremediation.</i>		
BTC4409	Bioteknologi Alam Sekitar / Environmental Biotechnology	3(2+1)
Prasyarat: BTC3401		
Kursus merangkumi penggunaan bioteknologi dalam mengawal dan memanipulasi pencemaran alam sekitar. Integrasi bioteknologi alam sekitar, isu terkini alam sekitar dan penyelesaian alternatif bagi pengawalan pencemaran alam sekitar dibincangkan.		
<i>This course encompasses the application of biotechnology in mitigating and manipulating the environmental pollutions. The integration in environmental biotechnology, current environmental issues and alternative solutions for maintaining the environmental pollution are discussed.</i>		
BTC4410	Biojisim dan Biorefinari / Biomass and Biorefinery	3(2+1)
Prasyarat: BTC3401		
Kursus ini memfokuskan kepada penukaran biojisim kepada produk bio menerusi konsep biorefinari. Jenis, ciri dan penggunaan biojisim, serta proses yang terlibat dalam sistem biorefinari diterangkan. Kajian kes melibatkan perbandingan pelbagai proses dan sistem biorefinari turut dibincangkan.		
<i>This course focuses on the conversion of biomass into bioproducts through biorefinery concept. Types, properties and utilization of biomass, as well as processes involved in a biorefinery system are explained. Case studies involving the comparison of different biorefinery processes and systems are also discussed</i>		
BTC4502	Pengurusan Biokeusahawanan / Bioentrepreneurship Management	3(1+2)
Prasyarat: BTC3502		
Kursus ini merangkumi pelbagai isu dalam pengurusan perniagaan seperti kepimpinan, kawalan kualiti, promosi, perundangan dan cabaran dalam syarikat start-up. Kaedah pitching turut diberi penekanan untuk menarik pelabur berpotensi bagi pembiayaan perniagaan.		
<i>This course encompasses issues in business management such as leadership, quality control, promotion, legislation and challenges in start-up company. The pitching method is also emphasized to attract potential investors for business financing.</i>		
BTC4503	Perniagaan Bioteknologi / Biotechnology Business	3(1+2)
Prasyarat: BTC4502		
Kursus ini merangkumi kaedah bagi pendaftaran perniagaan atau syarikat bioteknologi yang sah dari segi undang-undang. Proses penghasilan prototaip produk atau perkhidmatan bioteknologi bagi tujuan promosi dan juga potensi perkembangan perniagaan turut dibincangkan.		
<i>This course encompasses legal registration methods for biotechnology business or company. The process of producing prototype of biotechnology products or services for promotional purposes and potential business expansion are also discussed.</i>		

BTC4504	Pemindahan Teknologi Dalam Pengkomersialan Bioperniagaan / Technology Transfer in Commercialisation of Biobusiness	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi kepentingan konsep dan proses pemindahan teknologi dalam pengkomersialan perniagaan berasaskan bio. Perlindungan perundangan, komersialisasi harta intelek, etika dan kepentingan aset tidak ketara termasuk peluang, cabaran serta penyelesaian diberi penekanan.		
<i>This course encompasses the importance of concept and process of technology transfer in commercialising bio-based business. Legal protection, commercialisation of intellectual property, ethics and the importance of intangible assets including the opportunities, challenges as well as solutions are emphasized.</i>		

BTC4902	Latihan Industri / Industrial Training	12(0+12)
Prasyarat: Dengan kebenaran jabatan		
Kursus ini memperkenalkan pelajar kepada persekitaran kerja yang sebenar di industri/organisasi. Latihan merangkumi aplikasi aspek teori dan praktikal yang telah dipelajari dengan amalan semasa di tempat kerja. Kemahiran menyelesaikan masalah dan komunikasi turut diberi penekanan.		
<i>This course introduces students to real working environment in industries/organizations. Training includes application of the theoretical and practical aspects that have been studied with current practices in the workplace. Problem solving and communication skills are also emphasized.</i>		

BTC4904	Khidmat Pembelajaran dalam Bioteknologi / Service Learning in Biotechnology	1(0+1)
Prasyarat: BTC3001		
Kursus ini merangkumi penglibatan pelajar dalam aktiviti bersama komuniti / industri bagi meningkatkan kesedaran dan kefahaman orang awam berkenaan peranan bioteknologi dalam kehidupan seharian. Proses pembelajaran merangkumi perancangan aktiviti bersama komuniti, demonstrasi dan aktiviti 'hands-on' yang memenuhi keperluan komuniti.		
<i>This course encompasses involvement of students in community / industrial activities to promote awareness and understanding to the public on the role of biotechnology in their daily lives. The learning process involves activities planning with the community, demonstrations and hands-on activities to meet community needs.</i>		

BTC4959	Disertasi Bachelo / Bachelor Dissertation	6(0+6)
Prasyarat: Tiada		
Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui reka bentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.		
<i>This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.</i>		

BTC4991	Seminar / Seminar	1(0+1)
Prasyarat: BTC4959		
Kursus ini merangkumi penyediaan dan penyampaian seminar secara berkesan mengenai projek penyelidikan dalam bidang teknologi bioproses. Pendedahan kepada pelbagai teknik penyampaian seminar dicapai dengan menghadiri seminar tertentu di dalam kampus.		
<i>This course encompasses the effective preparation and delivery of seminar on research project in bioprocess technology. Exposure to different techniques of seminar presentation is achieved by attending selected seminars held in campus.</i>		

4. Jabatan Biologi Sel dan Molekul / Department of Cell & Molecular Biology

BSM3101	Biologi Sel dan Perkembangan / Cellular and Developmental Biology	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi prinsip biologi dan membandingkan pelbagai jenis sel seperti prokariot, eukariot dan virus, dari segi fisiologi dan organisasi genom. Konsep kitaran sel, struktur dan fungsi pelbagai jenis sel dan organel dibandingkan. Teori endosimbiosis, struktur dan fungsi membran sel dan nukleus serta mekanisme pengangkutan molekul juga diterangkan. Proses perkembangan dan perbezaan sel eukariot dan isu terkini berkaitan biologi sel dan perkembangan dibincangkan. <i>This course encompasses the principles of biology and compares various cell types such as prokaryotes, eukaryotes and viruses in terms of physiology and genome organisation. Concept of the cell cycle, structure and function of various cell types and organelles are compared. Endosymbiosis theory, structure and function of sel and nuclear membrane and the mechanism of molecular transport are explained. Cellular development and differentiation of eukaryotic cell and current issues related current issues related to cellular and developmental biology are discussed.</i>		
BSM3104	Prinsip Kultur Sel dan Tisu / Principles of Cell and Tissue Culture	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi prinsip dan prosedur asas kultur tisu tumbuhan dan haiwan. Keperluan asas untuk mewujudkan dan memelihara kultur sel di dalam makmal akan ditekankan. Ini termasuklah pelbagai kaedah seperti kultur embrio zigotik, kultur kalus dan organogenesis. Kestabilan genetik dan variasi somaklon akan dibincangkan. Kursus ini juga meliputi pelbagai konsep dan teknik dalam pengkulturan sel haiwan. Prinsip penghasilan antibodi monoklonal dan perbagai asai sitotoksik juga akan dibincangkan. <i>This course encompasses the principles and basic procedures of cell and tissue culture. The basic requirements for establishing and maintaining cell cultures in the laboratory will be emphasized. This includes various methods such as Zygotik embryo culture, callus culture and organogenesis. Genetic stability and somaclonal variation are discussed. This course also covers various concept and techniques in animal cell culture. Principles of monoclonal antibody production and various cytotoxic assays are also discussed.</i>		
BSM3201	Biologi Molekul / Molecular Biology	3(3+0)
Prasyarat: Tiada		
Kursus ini merangkumi prinsip biologi molekul seperti struktur dan peranan DNA dan RNA sebagai bahan genetik, struktur kromosom dan pewarisan di luar kromosom. Elemen yang boleh ditransposisikan, replikasi DNA, transkripsi, translasi, mutasi, metilasi, pembaikan DNA dan hubungan antara proses tersebut dibincangkan. Pengawalaturan ekspresi gen di antara prokariot dan eukariot dibandingkan. <i>This course encompasses the principles of molecular biology such as the structure and role of DNA and RNA as genetic materials, structure of chromosome and extrachromosomal inheritance. Transposable elements, DNA replication, transcription, translation, mutation, methylation, DNA repair and relationship between all these processes are discussed. The regulations of gene expression in eukaryotes and prokaryotes are compared.</i>		

BSM3202	Kejuruteraan Genetik / Genetic Engineering	4(3+1)
Prasyarat: BSM3201		
Kursus ini merangkumi teknik asas untuk pengklonan dan manipulasi gen. Prosedur pengklonan, penyaringan dan pengenalanpastian gen asing, vektor pengklonan, pembatasan vektor dan selitan DNA dan proses transformasi dibincangkan. Teknologi PCR dan penjujukan DNA juga diterangkan. Isu etika berkaitan dengan teknologi DNA rekombinan turut diberikan penekanan.		
<i>This course encompasses the basic techniques for cloning and manipulation of genes. Procedures for cloning, screening and identification of heterologous genes, cloning vectors, restriction of vector and insertion of DNA and transformation process are discussed. PCR technology and DNA sequencing are also explained. Ethical issues in relation to recombinant DNA technology are emphasised.</i>		

BSM3204	Prinsip Genetik / Principles of Genetics	3(3+0)
Prasyarat: Tiada		
Genetik, sains pewarisan dan variasi dibincangkan berdasarkan penemuan Mendel. Peranan dan aplikasi penting hukum genetik asas seperti Hukum Mendel dan Prinsip Hardy-Weinberg dalam ciri pewarisan di peringkat individu dan populasi serta relevannya hukum ini di dalam kehidupan manusia juga dibincangkan. Pelbagai konsep dan cabang genetik terkini turut diperkenalkan. Aplikasi konsep genetik dalam pemuliharaan biodiversiti dan biosumber dibincangkan.		
<i>Genetics, the science of heredity and variation are discussed based on Mendel's discoveries. The important roles and applications of basic genetic principles such as Mendel's Law and Hardy-Weinberg Principle in hereditary at the individual and population levels as well as their relevance in human life today are also discussed. Various recent concepts and branches of genetics are also introduced. Applications of genetic concepts in biodiversity and bioresource conservation are discussed.</i>		

BSM3205	Teknik Penyelidikan Dalam Biologi Molekul / Research Techniques in Molecular Biology	4(3+1)
Prasyarat: BSM3202		
Kursus ini merangkumi teori dan teknik asas penyelidikan biologi molekul dan kegunaannya dalam pelbagai bidang bioteknologi. Teknik lanjutan untuk analisis DNA, RNA dan protein termasuk penghasilan protein heterologus turut dibincangkan.		
<i>This course encompasses the theories and basic techniques used in molecular-biological research and its application in various fields. Advanced techniques in analysis of DNA, RNA and protein including heterologous protein production are also discussed.)</i>		

BSM3402	Imunologi Sel dan Molekul / Cell and Molecular Immunology	3(3+0)
Prasyarat: BSM3403		
Kursus ini merangkumi konsep penting dalam imunologi termasuk ciri asas tindakbalas imun, sel dan tisu dalam sistem imun serta mekanisme keimunan. Biologi limfosit T dan B termasuk perkembangan sel T dan B dari sel induk, pengaktifan dan pengawalaturan sel T dan B, antibodi dan antigen, molekul kompleks keserasian histo utama, imunologi transplantasi, hipersensitiviti dan autoimuniti dibincangkan.		
<i>This course encompasses the important concepts in immunology including basic properties of immune responses, cells and tissues in the immune system and the mechanism of immune responses. The biology of T and B lymphocytes including development of T and B cells from the progenitor cells, activation and regulation of T and B cells, antibody and antigen, and Major Histocompatibility Complex (MHC) molecule, transplantation immunology, hypersensitivity and autoimmunity are discussed.</i>		

BSM3403	Teknik Penyelidikan Dalam Biologi Sel / Research Techniques in Cell Biology	4(3+1)
Prasyarat: BSM3101 atau BSM3104		
Kursus ini merangkumi teknik asas biologi sel dari bioteknologi tumbuhan dan haiwan dalam kultur sel dan tisu, penghasilan tumbuhan transgenik dan pembiakbakaan molekul. Penekanan diberikan kepada penggunaan bioteknologi dalam penerokaan ilmu berkaitan biologi sel tumbuhan dan haiwan serta aplikasinya dalam penambahbaikan tumbuhan dan haiwan. Implikasi etika penggunaan teknik biologi sel dalam pertanian dibincangkan.		
<i>This course encompasses the basic techniques of plant and animal biotechnology in plant cell and tissue culture, development of transgenic plant and molecular breeding. Emphasis is given to the use of biotechnology in the exploration of knowledge related to plant and animal cell biology and its applications in plant and animal improvement. Ethical implications of the use of cell biology techniques in agriculture are discussed.)</i>		

BSM4102	Mikroteknik/ Microtechniques	3(2+1)
Prasyarat: BSM3101		
Kursus ini merangkumi prinsip dan kaedah asas dalam kajian histologi tumbuhan dan haiwan. Integrasi antara morfologi mikroskopi dengan fisiologi asas sel dan tisu juga diterangkan. Kemahiran menggunakan mikroskop cahaya dan teknik asas histologi juga diberi penekanan.		
<i>This course encompasses the basic principles and methods of histological studies in plant and animal. Integration of microscopic morphology with basic physiology of cells and tissues are also explained. Skills in using light microscopy and basic histological techniques are also emphasized.)</i>		

BSM4203	Diagnostik Molekul / Molecular Diagnostics	2(2+0)
Prasyarat: BSM3101		
Kursus ini merangkumi penemuan yang membawa kepada kewujudan bidang yang dipanggil diagnostik molekul. Konsep saintifik dalam revolusi diagnostik molekul diterangkan. Sitogenetik molekul, onkologi molekul dan pengesanan mikroorganisma penyebab penyakit berjangkit dikaji. Teknologi lanjutan seperti tatasusunan mikro genom manusia dan penggunaannya dalam diagnosis penyakit dibincangkan.		
<i>This course encompasses the discoveries that have created a field called molecular diagnostics. Scientific concepts in the revolution of molecular diagnostics are explained. Molecular cytogenetics, molecular oncology and detection of infectious disease causing microorganisms are examined. Advanced technology such as human genome microarray and its applications in disease diagnosis are discussed.</i>		

BSM4204	Genomik Fungsian / Functional Genomics	3(3+0)
Prasyarat: BSM3201 atau BMY4310		
Kursus ini merangkumi prinsip genomik fungsian dan kaedah penyelidikan yang terlibat dalam bidang yang berkaitan. Biologi molekul yang menentukan fungsi biologi gen dan produknya dengan menggunakan jujukan genom dibincangkan. Aplikasi kaedah berskala genom diperkenalkan. Topik semasa dalam fungsian genomik juga dibincangkan.		
<i>This course encompasses the principles of functional genomics and research approaches involved in related research. The molecular biology that determines the biological function of genes and their products by making use of genome sequences are discussed. The applications of genome-wide approaches are introduced. Current topics in functional genomics are also discussed.</i>		

BSM4205	Pengawalaturan Ekspresi Gen / <i>Regulation of Gene Expression</i>	2(2+0)
Prasyarat: BSM3201		
Kursus ini memberi tumpuan kepada pengawalaturan transkripsi gen dalam organisma eukariot yang meliputi mekanisme genetic dan epigenetik. Mekanisme epigenetik pengawalaturan gen semasa pembedaan sel yang berkembang secara normal dan tidak normal juga dibincangkan dengan contoh yang sesuai.		
<i>The course focuses on the regulation of gene transcription in eukaryotic organisms that include genetic and epigenetic mechanisms. Epigenetic mechanisms of gene regulation in the differentiation of cells that develop normally and abnormally are also discussed using appropriate examples.)</i>		
BSM4301	Bioinformatik / <i>Bioinformatics</i>	3(2+1)
Prasyarat: BSM3201 atau BCH3107		
Kursus ini merangkumi penelitian konsep asas dan kaedah dalam bioinformatik. Penekanan adalah kepada aspek teori dan praktik dalam analisis dan pengolahan jujukan asid nukleik dan protein, serta penggunaan perisian dalam analisis. Kegunaan bioinformatik dalam biologi molekul dibincangkan.		
<i>The course encompasses details on fundamental concepts and methods in bioinformatics. Emphasis is on the theory and practical aspects of analysis and manipulations of nucleic acid and amino acid sequences, and the use of software in the analyses. Applications of bioinformatics in molecular biology are discussed.</i>		
BSM4302	Penjjukan Generasi Seterusnya / <i>Next Generation Sequencing</i>	3(2+1)
Prasyarat: BSM4301		
Kursus ini merangkumi konsep asas dan kaedah dalam teknologi penjjukan generasi seterusnya. Teori dan praktik ditekankan dalam analisis pelbagai data penjjukan menggunakan perisian bioinformatik yang sesuai. Aplikasi penjjukan generasi seterusnya juga dibincangkan.		
<i>The course encompasses the fundamental concepts and methods in next generation sequencing technologies. Theory and practical are emphasised on analysis of various sequencing data using appropriate bioinformatics tools. Applications of next generation sequencing technologies are also discussed.)</i>		
BSM4401	Kultur Sel dan Tisu Haiwan Lanjutan / <i>Advanced Animal Cell and Tissue Culture</i>	3(3+0)
Prasyarat: BSM3403		
Kursus ini merangkumi maklumat latar belakang penting untuk memungkinkan penggunaan teknik kultur sel haiwan sebagai pilihan untuk operasi pengeluaran protein yang diinginkan secara besar-besaran, terutamanya dalam penjagaan kesihatan. Teknik penting dalam memanipulasi dan merekabentuk kultur sel mamalia, terutamanya untuk biosintesis bahan biologi seperti protein rekombinan atau antibodi monoklonal dibincangkan. Prinsip dan teknik di sebalik manipulasi sel untuk penghasilan titisan sel berskala besar yang sangat produktif diterangkan.		
<i>This course encompasses the essential background information to enable the establishment of animal cell cultures as an option for large-scale production operations of desired proteins, mainly in healthcare. Important techniques in manipulating and designing mammalian cell cultures, particularly for the biosynthesis of biological substances such as recombinant proteins or monoclonal antibodies are discussed. The principles and techniques behind cell manipulations for scale-up production of highly productive cell lines are explained.</i>		

BSM4501	Kultur Sel dan Tisu Tumbuhan Gunaan / <i>Applied Plant Cell and Tissue Culture</i>	4(3+1)
Prasyarat: BSM3403		
Kursus ini merangkumi konsep penting bagi pembiakan klon tumbuhan pertanian, hortikultur dan perubatan. Kaedah bagi pengeluaran tumbuhan bebas dari patogen, variasi rintang penyakit dan tekanan serta kepentingannya dalam penghasilan varieti baru dibincangkan. Prinsip penyimpanan germplasma dan penghasilan metabolit sekunder serta potensi kejuruteraan genetik tumbuhan diterangkan.		
<i>This course encompasses the essential concepts for clonal propagation of agricultural, horticultural and medicinal plants. Methods for production of pathogen-free plants, disease-resistant and stress-tolerant strains and the importance in the production of new varieties are discussed. The principles of germplasm storage, secondary metabolites and potential of genetic engineering in plants are described.</i>		
BSM4503	Biologi Molekul Perkembangan Tumbuhan/ <i>Molecular Biology of Plant Development</i>	4(4+0)
Prasyarat: BSM3201 dan BSM3101		
Kursus ini merangkumi prinsip asas biologi perkembangan tumbuhan. Aspek perkembangan sel dan organ tumbuhan seperti struktur dan fisiologi sel, proses perkembangan sel dan organ dibincangkan. Hubungkait di antara pengekspresan gen dengan fungsi struktur dan fisiologi tisu tumbuhan diterangkan.		
<i>This course encompasses the basic principles of plant developmental biology. Aspects of plant cell and organ development, such as cell structures and physiology, cell and organ developmental processes are discussed. The relationship of gene expression to structure and physiological functions of plant tissues is explained.</i>		
BSM4504	Teknologi Tumbuhan Transgenik / <i>Transgenic Plant Technology</i>	4(3+1)
Prasyarat: BSM3202		
Kursus ini merangkumi aspek manipulasi dan penganalisaan genom tumbuhan. Teknik biologi molekul dalam pembiakbakaan tumbuhan diterangkan. Pelbagai isu berkaitan dengan produk terubahsuai secara genetik juga dibincangkan.		
<i>This course encompasses aspects of manipulation and analyses of plant genome. Applications of molecular biology techniques in plant breeding are explained. Various issues on genetically modified products are also discussed).</i>		
BSM4601	Kejuruteraan Protein / <i>Protein Engineering</i>	4(3+1)
Prasyarat: BSM3202 atau BMY4310		
Kursus ini merangkumi penggunaan teknik kimia dan genetik untuk mengubahsuai protein. Penekanan diberikan kepada teori penstabilan protein, teknik penulenan, analisis, ramalan struktur 3D dan teknik pengubahsuaian protein.		
<i>This course encompasses the use of genetic and chemical techniques to modify protein. Emphasis is on the theories related to protein stabilization, purification techniques, analysis, 3D structure prediction and protein modification techniques.</i>		
BSM4602	Proteomik / <i>Proteomics</i>	4(3+1)
Prasyarat: BSM3201 atau BMY4310		
Kursus ini merangkumi konsep, teknologi, dan aplikasi proteomik. Topik utama yang dibincangkan merangkumi kedinamikan dan kekompleksan proteom, pengubahsuaian dan kepelbagaian protein, ekspresi dan interaksi proteom, teknologi proteom, dan informatic protein. Aplikasi proteomik dalam pelbagai sektor penyelidikan juga dibincangkan.		

This course encompasses the concepts, technologies, and applications of proteomics. Major topics discussed include proteome dynamics and complexity, protein modifications and diversity, proteome expression and interaction, proteome technologies, and protein informatics. Applications of proteomics in various sectors of research are also discussed.)

BSM4603	Biologi Struktur / Structural Biology	3(3+0)
Prasyarat: BSM3201 atau BMY4310		
Kursus ini merangkumi pengetahuan asas mengenai struktur tiga dimensi protein. Kepentingan struktur protein dalam penentuan fungsi protein dibincangkan. Selain itu, beberapa teknik penting penentuan struktur protein secara kristalografi dan Nuclear Magnetic Resonance (NMR) juga diterangkan. Aplikasi berkaitan biologi struktur dalam bidang sains turut dibincangkan.		
<i>This course encompasses the basic knowledge about the three-dimensional structure of proteins. The importances of protein structures in determining the functions of proteins are discussed. In addition, a few major techniques in determining protein structures via crystallography and Nuclear Magnetic Resonance (NMR) are explained. The applications related to structural biology in other fields of science are also discussed.</i>		

BSM4701	Bahan Nano Dan Bioteknologi / Nanomaterials and Biotechnology	3(3+0)
Prasyarat: BSM3201 atau BCH3107		
Kursus ini merangkumi prinsip asas penggunaan bahan nano dalam kajian bioteknologi. Konsep sintesis dan pencirian bahan nano, serta instrumentasi dalam nanobioteknologi dibandingkan dan diterangkan. Kadar ketoksikan dan kelangsungan <i>in vitro</i> serta <i>in vivo</i> pelbagai bahan nano turut dibincangkan. Aplikasi bahan nano dalam pelbagai bidang seperti perubatan, pertanian, dan alam sekitar diterokai.		
<i>This course encompasses the basic principles of nano materials in biotechnological research. The concepts in the synthesis and characterization of nanomaterials, as well as the various instrumentation in nanobioteknologi are compared and described. The <i>in vitro</i> and <i>in vivo</i> toxicity and persistence of nano materials are also discussed. The applications of nano materials in various fields including medicine, agriculture, and the environment are explored.</i>		

BSM4902	Latihan Industri / Industrial Training	12(0+12)
Prasyarat: Dengan kebenaran Jabatan		
Kursus ini mendedahkan pelajar kepada persekitaran kerja yang sebenar di industri/organisasi. Latihan dijalankan di agensi kerajaan atau swasta selama 24 minggu. Latihan merangkumi aplikasi aspek teori dan praktikal yang telah dipelajari dengan amalan semasa di tempat kerja. Kemahiran menyelesaikan masalah dan komunikasi turut diberi penekanan.		
<i>This course introduces students to real working environment in industries/organizations. This training is conducted in government or private agencies for 24 months. Training includes application of the theoretical and practical aspects that have been studied with current practices in the workplace. Problem solving and communication skills are also emphasized.)</i>		

BSM4904	Pembelajaran Perkhidmatan dalam Biologi Sel dan Molekul/ Service Learning in Cell and Molecular Biology	1(0+1)
Prasyarat: BSM3202		
Kursus ini merangkumi aktiviti bagi meningkatkan kesedaran dan kefahaman komuniti tentang kepentingan biologi sel dan molekul dalam kehidupan harian. Proses pembelajaran dan bekerjasama dengan masyarakat luar bagi membina pengetahuan sains, demonstrasi dan aktiviti "hands-on" bagi memenuhi keperluan komuniti.		
<i>This course encompasses activities to increase awareness of community and to enhance understanding of the importance of cell and molecular biology in daily lives. The learning process involves working together with the public to develop science lessons, demonstrations and hands-on activities to meet community needs.</i>		

BSM4959	Disertasi Bachelo / Bachelor Dissertation	6(0+6)
Prasyarat: Tiada		
Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui rekabentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.		
<i>This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.</i>		

BSM4991	Seminar / Seminar	1(0+1)
Prasyarat: BSM4904		
Kursus ini merangkumi penyediaan dan penyampaian seminar secara berkesan mengenai projek penyelidikan dalam bidang biologi sel dan molekul. Pelajar perlu membuat sorotan literatur, menyusun dan membentangkan maklumat biologi sel dan molekul dalam satu seminar.		
<i>This course encompasses the effective preparation and delivery of seminars on research project in cell and molecular biology. Students are required to review the literature, organise and present information on cell and molecular biology in a seminar.</i>		